

SHIVANSH BAJAJ

AI/ML Engineering Student • Python Developer • Computer Vision

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PROFESSIONAL SUMMARY

Second-year B.Tech Computer Science student specializing in AI & ML, focused on building practical machine learning and computer vision systems rather than isolated tutorials. Shipped a real-time hand-gesture control system using MediaPipe and OpenCV with a custom stabilization layer and automated test coverage. Comfortable across the Python data-science stack (scikit-learn, Pandas, NumPy) and actively building AWS cloud fundamentals to take models beyond a local script. Seeking an AI/ML or Python development internship to apply this on production-scale problems.

TECHNICAL SKILLS

Languages: Python, Java, C, C++

ML / Data: scikit-learn, Pandas, NumPy, TensorFlow (learning), model evaluation, PCA

Computer Vision: OpenCV, MediaPipe, real-time gesture recognition

Cloud & Tools: AWS fundamentals, Git, GitHub, Linux/Ubuntu, VS Code

Core CS: Data Structures & Algorithms, OOP, unit testing (pytest)

PROJECTS

AI Gesture Control for Google Slides — Flagship Project

github.com/Shivanshbajaj1/AI-Gesture-Control-Slides

- Built a real-time computer vision system that controls Google Slides presentations using hand gestures, processing live webcam input with MediaPipe and OpenCV.
- Designed a custom majority-vote gesture stabilizer (6-of-8 frame buffer) to eliminate single-frame misfires and flicker, materially improving real-world reliability over a naive frame-by-frame classifier.
- Structured the codebase into four modules (gesture_utils, main, setup_models, README) with a 9-test pytest suite, reflecting production coding practices rather than notebook-only development.

Python PBL — End-to-End ML Pipeline

github.com/Shivanshbajaj1/Python_PBL

- Built a complete machine learning pipeline covering data preprocessing, exploratory data analysis, PCA-based dimensionality reduction, and classification on real datasets.
- Applied systematic debugging and performance tuning across the pipeline, from data cleaning through model evaluation.

Typing Analysis & Accuracy Tracking Platform

In progress

- Building a Monkeytype-inspired typing platform with per-key accuracy tracking and a responsive TypeScript frontend, extending practical software development skills beyond Python.

EDUCATION

Bachelor of Technology (B.Tech) — Computer Science, Specialization in AI & ML

2024 – 2028

Graphic Era Deemed To Be University, Dehradun

STRENGTHS

Quick Learner • Analytical Thinking • Problem Solving • Team Collaboration